

Cost-Effective Predictive Modeling for Customer Targeting

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Task: rank 5 000 customers by acceptance probability; contact top $k \leq 1000$.

Score:

$$\text{Score} = 10 \text{ TP} - 5 \text{ FP} - 200 N_{\text{vars}}$$

- 5 000 training rows, 500 features, $\approx 50\%$ positive rate
- each declared feature costs 200 pts (≈ 20 extra TPs to break even)
- break-even contact: $E[\text{contact}] = 15p - 5 > 0 \Rightarrow p > 1/3$

- 1 **Feature selection** — 500 → ranked top_01–top_26
- 2 **Baselines** — no-skill floors, feature-tax demo (optional)
- 3 **Modelling** — five targeting strategies
- 4 **Comparison** — OOF business score across approaches

Strategies:

- Top- k profit curve
- EV profit targeting
- Rank fusion committee
- Cluster split (KMeans)
- Segment-aware (GMM)

Three stages

- **Stage 1** (500→80): variance, MI, LightGBM — combined rank
- **Stage 2** (80→26): correlation, VIF, LASSO path
- **Stage 3**: drop-column CV ranking by *business score*

Also compared (diagnostic): MDI, permutation importance, Boruta — agree on top features, but drop-column CV used for modelling.

Top features	Rank	Feature
	1	var_389
	2	var_308
	3	var_93
	4	var_87
	5	var_254

CV business by subset	
Subset	Business
top_01	2 301
top_03	1 891
top_05	1 490
top_08	898
top_10	486
top_26	−2 684

Reference models on full train (5-fold CV, top- k business score with penalty).

Baseline	Feat.	Business	F1@top- k
Prior / uniform dummy	0	2 475	0.666
Stratified dummy	0	2 474	0.665
Random ranker (mean)	0	$\approx$ 2 475	0.665
Logistic (var_389)	1	2 304	0.668
Logistic (500 feat.)	500	-97 493	0.668

Raw features vs. PCA

Logistic regression: first k ranked raw features vs. first k PCA components (500 vars scaled).

k	F1 raw	F1 PCA	AUC raw	AUC PCA
1	0.668	0.666	0.555	0.555
5	0.669	0.665	0.581	0.591
10	0.669	0.666	0.594	0.598
25	0.670	0.667	0.603	0.598

Raw ranked features win on F1 at every k ; might indicate that PCA captures noise.

Approach	Idea
Top- k curve	Compare LR / LGBM / XGB; $k^* = \arg \max_k \text{Score}_k$ on OOF curve
EV targeting	Isotonic calibration; $k = \min(k_{\text{ev}}, k_{\text{elbow}}, k_{\text{nested}})$
Rank fusion	4 experts, OOF-weighted $\hat{p}$; 8-feature union penalty
Cluster split	KMeans ($K=4$) + per-cluster logistic on same top_*
Segment-aware	GMM on top_10 (latent); logistic on top_03 per segment

Results: OOF comparison

Approach	Feat.	k	TP	FP	OOF
Segment targeting	3	998	598	400	3 380
Top- k profit curve	1	997	565	432	3 290
Cluster split	5	998	618	380	3 280
EV targeting	1	939	536	403	3 145
Rank fusion	8	994	577	417	2 085
Prior dummy (ref.)	0	1 000	–		$\approx$ 2 475

Best classic config: GMM segments + per-segment logistic on top_03, $k=998$.

- **Difficult/noisy data** — model on entire dataset did not improve F1 (PCA analysis).
- **Feature tax dominates** — var_389 alone wins CV; more vars ↓ business score
- **Local rankers help** — cluster split (3 280) → segment targeting (3 380)

Thank You

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